

Real Estate Investment Advisor– Insight Report

Introduction

Real estate investment is one of the most significant financial decisions for individuals as well as institutions. The rapid growth of urbanization, population, and infrastructure development has made property investment more complex and data-intensive. Traditional decision-making methods rely heavily on intuition and limited market knowledge, which may lead to inaccurate investment choices.

With the increasing availability of real estate data, data analytics and machine learning techniques can be used to extract meaningful insights and identify profitable investment opportunities. This project focuses on analyzing real estate data to determine whether a property is a **good investment** based on multiple factors such as location, property type, area, price trends, crime rate, and infrastructure facilities.

Real Estate Investment Analysis

Step 1: Data Preprocessing

A. Handling Missing Values and Duplicates

Raw real estate datasets often contain missing entries due to incomplete surveys or data collection errors. Missing values were handled as follows:

- **Numerical attributes** (Price, Area, Crime Rate, Infrastructure Score) were imputed using the **mean or median**, depending on skewness.
- **Categorical attributes** (Location, Property_Type) were filled using the **most frequent value (mode)**.
- Duplicate property records were identified based on **Location, Area, BHK, and Price** and removed to prevent bias in analysis.

This step ensured improved data quality and reliable insights.

B. Feature Scaling and Normalization

To ensure all numerical features contribute equally to model performance:

- **Min-Max Scaling** was applied to attributes such as Price, Area, Crime Rate, and Infrastructure Score.
- Scaling helped algorithms converge faster and improved interpretability in Exploratory Data Analysis (EDA).

C. Encoding Categorical Features

Machine learning models require numerical inputs. Hence:

- **Location** and **Property_Type** were encoded using:

- Label Encoding for ordinal data
 - One-Hot Encoding for nominal data
- This prevented model bias caused by arbitrary numerical assignments.

D. Feature Engineering

New features were created to enhance predictive power:

- **Price per Sqft** = Property Price / Area
- **School Density Score** based on the number of schools within a specific radius
- **Infrastructure Index** combining transport access, hospitals, and public facilities

These engineered features helped capture real-world investment dynamics.

E. Target Variable Creation – “Good Investment”

A binary label **Good Investment** was created using domain knowledge:

- Properties with an **appreciation rate greater than a defined threshold** (e.g., 10%) were labeled **1 (Good Investment)**.
- Others were labeled **0 (Not a Good Investment)**.

This enabled classification-based analysis and decision-making.

Step 2: Exploratory Data Analysis (EDA)

1. Price Trends Across Cities

Analysis revealed:

- Metro cities showed **higher average property prices**.
- Tier-2 cities exhibited faster appreciation due to upcoming infrastructure developments.

Visualizations such as bar charts and line graphs were used for better comparison.

2. Correlation Between Area and Investment Return

EDA showed:

- Moderate positive correlation between **property area and investment return**.
- Larger properties generally yield higher resale value but require higher initial investment.

A correlation heatmap helped visualize these relationships.

3. Impact of Crime Rate on Investment Quality

- Higher crime rates were associated with **lower Good Investment classification**.

- Safe neighborhoods demonstrated stronger long-term appreciation.

This insight highlights the importance of safety indicators in real estate decisions.

4. Infrastructure Score vs Resale Value

- Properties with higher infrastructure scores (metro connectivity, hospitals, malls) showed:
 - Higher resale demand
 - Faster appreciation

Scatter plots confirmed a strong positive relationship.

Step 5: Streamlit-Based Web Application

1. Application Overview

A **Streamlit dashboard** was developed to provide an interactive user experience for property investment analysis.

2. User Input Form

The application allows users to input:

- Location
- Property Type
- Area (Sqft)
- Budget / Price Range
- Number of BHK
- Infrastructure Score

These inputs help customize recommendations.

3. Property Filtering Options

Users can filter properties based on:

- Area range
- Price range
- BHK count
- Location

This helps narrow down properties suitable for individual investment goals.

4. Investment Prediction Output

Based on user input:

- The system predicts whether a property is a **Good Investment or Not**
- Displays key insights such as:
 - Expected appreciation
 - Price per Sqft
 - Risk indicators (crime rate, infrastructure)

Conclusion

This project demonstrates how data preprocessing, feature engineering, and exploratory data analysis can effectively support real estate investment decisions. By cleaning and transforming raw data, meaningful patterns and relationships were identified. Key factors such as location, infrastructure, crime rate, and area were shown to significantly influence investment quality. The creation of the “Good Investment” label enabled clear classification of properties. The Streamlit application provided an interactive and user-friendly platform for analysis. Overall, the system helps investors make informed, data-driven decisions. This approach reduces risk and improves the accuracy of property investment evaluation.